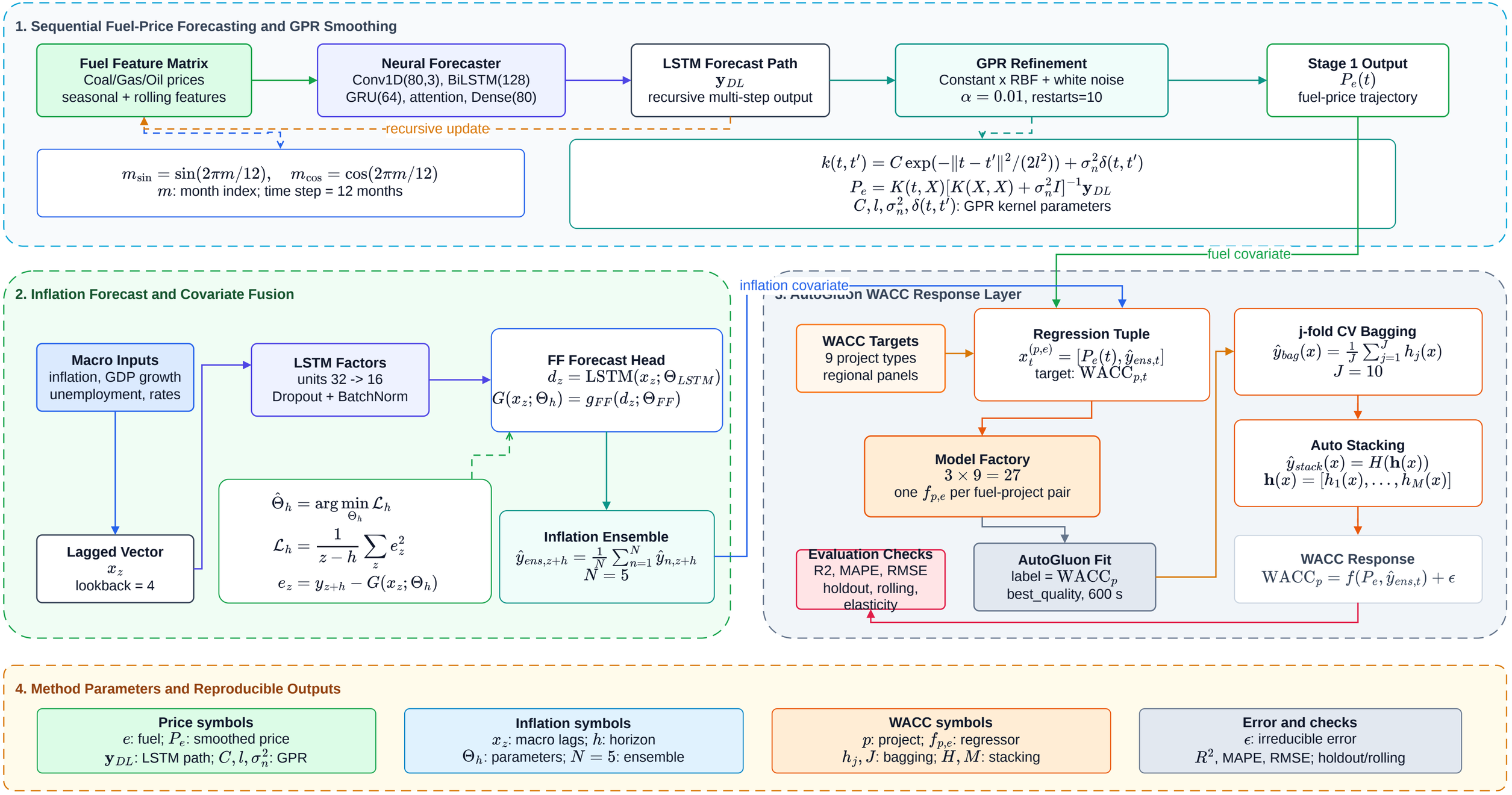

1. Sequential Fuel-Price Forecasting and GPR Smoothing
Fuel Feature Matrix
Coal/Gas/Oil prices
seasonal + rolling features
Neural Forecaster
Conv1D(80,3), BiLSTM(128)
GRU(64), attention, Dense(80)
LSTM Forecast Path
$\mathbf{y}_{DL}$
recursive multi-step output
GPR Refinement
Constant x RBF + white noise
$\alpha = 0.01$, restarts=10
Stage 1 Output
$P_e(t)$
fuel-price trajectory
recursive update
$m_{\sin} = \sin(2\pi m/12), \quad m_{\cos} = \cos(2\pi m/12)$
m: month index; time step = 12 months
$k(t,t') = C\exp(-\|t-t'\|^2/(2l^2)) + \sigma_n^2\delta(t,t')$
$P_e = K(t,X)[K(X,X)+\sigma_n^2 I]^{-1}\mathbf{y}_{DL}$
$C, l, \sigma_n^2, \delta(t,t')$: GPR kernel parameters
fuel covariate
2. Inflation Forecast and Covariate Fusion
Macro Inputs
inflation, GDP growth
unemployment, rates
LSTM Factors
units 32 -> 16
Dropout + BatchNorm
FF Forecast Head
$d_z = \mathrm{LSTM}(x_z;\Theta_{LSTM})$
$G(x_z;\Theta_h) = g_{FF}(d_z;\Theta_{FF})$
Lagged Vector
x_z
lookback = 4
$\hat{\Theta}_h = \arg\min_{\Theta_h} \mathcal{L}_h$
$\mathcal{L}_h = \frac{1}{z-h}\sum_z e_z^2$
$e_z = y_{z+h} - G(x_z;\Theta_h)$
Inflation Ensemble
$\hat{y}_{ens,z+h} = \frac{1}{N}\sum_{n=1}^{N}\hat{y}_{n,z+h}$
$N = 5$
inflation covariate
3. AutoGluon WACC Response Layer
WACC Targets
9 project types
regional panels
Regression Tuple
$x_t^{(p,e)} = [P_e(t), \hat{y}_{ens,t}]$
target: $\mathrm{WACC}_{p,t}$
j-fold CV Bagging
$\hat{y}_{bag}(x) = \frac{1}{J}\sum_{j=1}^{J} h_j(x)$
$J = 10$
Model Factory
$3 \times 9 = 27$
one $f_{p,e}$ per fuel-project pair
Auto Stacking
$\hat{y}_{stack}(x) = H(\mathbf{h}(x))$
$\mathbf{h}(x) = [h_1(x), \ldots, h_M(x)]$
Evaluation Checks
R2, MAPE, RMSE
holdout, rolling,
elasticity
AutoGluon Fit
label = WACC_p
best_quality, 600 s
WACC Response
$\mathrm{WACC}_p = f(P_e, \hat{y}_{ens,t}) + \epsilon$
4. Method Parameters and Reproducible Outputs
Price symbols
e: fuel; P_e: smoothed price
$\mathbf{y}_{DL}$: LSTM path; C, l, σ_n^2: GPR
Inflation symbols
x_z: macro lags; h: horizon
Θ_h: parameters; $N = 5$: ensemble
WACC symbols
p: project; $f_{p,e}$: regressor
h_j, J: bagging; H, M: stacking
Error and checks
ϵ: irreducible error
R^2, MAPE, RMSE; holdout/rolling